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EX-2 Answers

1) Given

$$P(X=i, Y=j) = p_{ij}$$

$$X \in \{1, 2\}$$

$$Y \in \{1, 2, 3\}$$

All $p_{ij} > 0$.a) Calculate marginal distribution, $\text{Prob}(X=2)$ & $\text{Prob}(Y=1)$

$$P(X=2) = \sum_{j=1}^3 P(X=2, Y=j) = p_{21} + p_{22} + p_{23}$$

$$P(Y=1) = \sum_{i=1}^2 P(X=i, Y=1) = p_{11} + p_{21}$$

b) Calculate $\text{Prob}(Y=1 | X=2)$

By def of Conditional Prob,

$$P(Y=1 | X=2) = \frac{P(X=2, Y=1)}{P(X=2)}$$

$$\text{We know } P(X=2, Y=1) = p_{21}$$

$$P(X=2) = p_{21} + p_{22} + p_{23}$$

$$\Rightarrow P(Y=1 | X=2) = \frac{p_{21}}{p_{21} + p_{22} + p_{23}}$$

c) Calculate $\text{Prob}(X < Y)$ So, we want to sum all joint probabilities where $i < j$ The combinations which satisfies $i < j$

$$\rightarrow (1, 2) \Rightarrow p_{12}$$

$$\rightarrow (2, 3) \Rightarrow p_{23}$$

$$\rightarrow (1, 3) \Rightarrow p_{13}$$

$$\Rightarrow P(X < Y) = p_{12} + p_{23} + p_{13}$$



2) a) Given

Independent observation $\{K_i, x_i\}_{i=1}^n$

Each $x_i \sim \text{Gamma}(K_i, \theta)$

For each i , the density is,

$$f(x_i | K_i, \theta) = \frac{\theta^{K_i}}{\Gamma(K_i)} x_i^{K_i-1} e^{-\theta x_i}$$

As the data points are independent, the joint likelihood for n observations is the product of individual densities.

$$L(\theta) = \prod_{i=1}^n f(x_i | K_i, \theta) = \prod_{i=1}^n \left(\frac{\theta^{K_i}}{\Gamma(K_i)} x_i^{K_i-1} e^{-\theta x_i} \right)$$

$$\Rightarrow L(\theta) = \theta^{\sum K_i} \cdot \left(\prod_{i=1}^n \frac{x_i^{K_i-1}}{\Gamma(K_i)} \right) \cdot e^{-\theta \sum x_i}$$

b) We know the value of $L(\theta)$ from the previous derivation,

Only θ varies, all others are constants, so let's take log likelihood.

$$\log L(\theta) = \sum K_i \log \theta - \theta \sum x_i + (\text{constants w.r.t } \theta)$$

As we are maximizing, we can ignore constants.

$\Rightarrow$ Let us differentiate & set to zero.

$$\frac{d}{d\theta} \log L(\theta) = \frac{\sum K_i}{\theta} - \sum x_i = 0$$

$$\Rightarrow \frac{\sum K_i}{\theta} = \sum x_i \Rightarrow \theta = \frac{\sum K_i}{\sum x_i}$$

So, by maximizing θ with MLE,

$$\theta = \frac{\sum_{i=1}^n K_i}{\sum_{i=1}^n x_i}$$

c) Given Prior of θ is Gamma(k_0, x_0)

$$\rightarrow p(\theta) = \frac{\theta^{k_0-1} e^{-\theta x_0}}{\Gamma(k_0) x_0^{k_0}}$$

We know

$$p(\theta | \sum_{i=1}^n K_i, x_i) \propto \text{Likelihood} \times \text{Prior}$$

$$\text{So, } p(\theta | \sum_{i=1}^n K_i, x_i) \propto \theta \sum_{i=1}^n K_i e^{-\theta x_i} \cdot \theta^{k_0-1} e^{-\theta x_0}$$

$\Rightarrow$ Posterior is,

$$\theta | (K_i, x_i)_{i=1}^n \sim \text{Gamma}(k_0 + \sum_{i=1}^n K_i, x_0 + \sum_{i=1}^n x_i)$$

$$\Rightarrow x_{\text{post}} = x_0 + \sum_{i=1}^n x_i ; k_{\text{post}} = k_0 + \sum_{i=1}^n K_i$$

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3) Given

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \end{bmatrix} \sim N \left(\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}, \begin{bmatrix} 3 & 0 & -1 \\ 0 & 1 & -1 \\ -1 & -1 & 2 \end{bmatrix} \right)$$

a) For two variables X_i & X_j to be independent with each other, their covariance

$\Sigma_{i,j}$ needs to be zero.

$$\text{Here } \text{Cov}(X_1, X_2) = 0$$

$$\text{Cov}(X_1, X_3) = -1$$

$$\text{Cov}(X_2, X_3) = -1$$

So, the Independent pair is X_1 & X_2

b) Given

$$Z = X_1 - a X_2 + b X_3$$

A linear combination of multivariate normal variables are independent of one of them

Here if Z is independent of X_3 , if & only if,

$$\text{Cov}(Z, X_3) = 0$$

$$\text{So, } \text{Cov}(Z, X_3) = \text{Cov}(X_1 - aX_2 + bX_3, X_3) = 0$$

$$\Rightarrow \text{Cov}(Z, X_3) = \text{Cov}(X_1, X_3) - a \cdot \text{Cov}(X_2, X_3) + b \cdot \text{Var}(X_3)$$

We know,

$$\text{Cov}(X_1, X_3) = -1$$

$$\text{Cov}(X_2, X_3) = -1$$

$$\text{Var}(X_3) = 2$$

$$\Rightarrow \text{Cov}(Z, X_3) = -1 - a(-1) + 2b = 0$$

$$\Rightarrow -1 + a + 2b = 0 \Rightarrow a + 2b = 1$$

So, it is possible to set values of a & b such that Z is independent with X_3 .

An example, Let $a = 1$,

$$\Rightarrow 1 + 2b = 1 \Rightarrow b = 0$$

$$\Rightarrow Z = X_1 - 1 \cdot X_2 + 0 \cdot X_3 = X_1 - X_2$$

So Z is independent of X_3

$$c) \text{ We know } Z = X_1 - aX_2 + bX_3$$

We need to find whether

$$\text{Cov}(Z, X_2 | X_1 = x_1) = 0 \text{ is possible where } x_i \in \mathbb{R}.$$

Using the rule of conditional covariance,

$$\text{Cov}(Z, X_2 | X_1 = x_1) = \text{Cov}(Z, X_2) - \frac{\text{Cov}(Z, X_1) \cdot \text{Cov}(X_1, X_2)}{\text{Var}(X_1)}$$

We know all the values here from covariance matrix

$$\text{Now } \text{Cov}(Z, X_2) = \text{Cov}(X_1, X_2) - a \cdot \text{Cov}(X_2, X_2) + b \cdot \text{Cov}(X_3, X_2)$$

$$= 0 - a + b \Rightarrow -a + b$$

$$\text{Cov}(Z, X_1) = \text{Cov}(X_1, X_1) - a \cdot \text{Cov}(X_2, X_1) + b \cdot \text{Cov}(X_3, X_1)$$

$$= 3 - a(0) + b(-1) = 3 - b$$



$$\Rightarrow C_v(z, x_2 | x_1) = (-a-b) - \frac{(3-b)(0)}{3} = -a-b$$

$$\Rightarrow -a-b=0 \Rightarrow a=-b$$

So, it is possible.

$$z = x_1 - a x_2 + b x_3 = x_1 - (-b) x_2 + b x_3$$

$$= x_1 + b(x_2 + x_3)$$

$$\text{So, if } a=2 \Rightarrow b=-2$$

$$\Rightarrow z = x_1 - 2(x_2 + x_3)$$

This is conditionally independent of x_2 given x_1 .

4) Given

$$\begin{aligned} x^{(1)} &= 4 \\ x^{(2)} &= 10 \\ x^{(3)} &= 16 \\ x^{(4)} &= 20 \\ x^{(5)} &= 26 \end{aligned}$$

a) Initial centroids,

$$\mu_1 = 2, \mu_2 = 3, \mu_3 = 4$$

Iteration 1: Distance to each centroid

Point	μ_1	μ_2	μ_3	Closest
4	2	1	0	μ_3
10	8	7	6	μ_3
16	14	13	12	μ_3
20	18	17	16	μ_3
26	24	23	22	μ_3

All points go to ($\mu_3=4$) cluster 3

Now update centroid of cluster 3.

$$\mu_3 = \frac{4+10+16+20+26}{5} = 15.2$$

$$\Rightarrow \mu_1 = 2, \mu_2 = 3, \mu_3 = 15.2$$

Iteration 2: Distance to each centroid

Point	$\mu_1=2$	$\mu_2=3$	$\mu_3=18.2$	Closest
4	2	1	11.2	μ_2
10	8	7	5.2	μ_3
16	14	13	0.8	μ_3
20	18	17	4.8	μ_3
26	24	23	10.8	μ_3

$\Rightarrow$ 4 goes to cluster 2,
Rest all goes to cluster 3.

$$\Rightarrow \mu_2 = 4$$

$$\mu_3 = \frac{10+16+20+26}{4} = \frac{72}{4} = 18$$

$$\Rightarrow \mu_1 = 2, \mu_2 = 4, \mu_3 = 18$$

It 3:

Point	$\mu_1=2$	$\mu_2=4$	$\mu_3=18$	Closest
4	2	0	14	μ_2
10	8	6	8	μ_2
16	14	12	2	μ_3
20	18	16	2	μ_3
26	24	22	8	μ_3



$$\Rightarrow \mu_2 = \frac{4+10}{2} = 7$$

$$\mu_3 = \frac{16+20+26}{3} = \frac{62}{3} = 20.67$$

$$\mu_1 = 2$$

It 4:

Point	$\mu_1=2$	$\mu_2=7$	$\mu_3=20.67$	Closest
4	2	3	16.67	μ_1
10	8	3	10.67	μ_2
16	14	9	4.67	μ_3
20	18	13	0.67	μ_3
26	24	19	5.33	μ_3

$$\Rightarrow \mu_1 = 4$$

$$\mu_2 = 10$$

$$\mu_3 = \frac{16+20+26}{3} = 20.67 \text{ (Same)}$$

It 5:

Point	$\mu_1=4$	$\mu_2=10$	$\mu_3=20.67$	Closest
4	0	3.36	16.67	μ_1
10	6	0	10.67	μ_2
16	12	6	4.67	μ_3
20	16	10	0.67	μ_3
26	22	16	5.33	μ_3



⇒ Convergence Achieved ✓

Final centroids:

$$\mu_1 = 4, \mu_2 = 10, \mu_3 = 20.67$$

b) No, K-means converges to a local minimum, not always global minimum.

So, it heavily depends on initial centroid positions.

Ex: Let us initialize $\mu_1 = 4, \mu_2 = 16, \mu_3 = 26$

It 1:

Point	$\mu_1 = 4$	$\mu_2 = 16$	$\mu_3 = 26$	Closest
4	0	12	22	μ_1
10	6	6	16	μ_1 (tie, choose μ_1)
16	12	0	10	μ_2
20	16	4	6	μ_2
26	22	10	0	μ_3

$$\Rightarrow \mu_1 = \frac{4+10}{2} = 7$$

$$\mu_2 = \frac{16+20}{2} = 18$$

$$\mu_3 = 26$$

It 2:

Point	$\mu_1 = 7$	$\mu_2 = 18$	$\mu_3 = 26$	Closest
4	3	14	22	μ_1
10	3	8	16	μ_1
16	9	2	10	μ_2
20	13	2	6	μ_2
26	19	8	0	μ_3

⇒ Convergence Achieved ✓

Final Centroids:

$$\mu_1 = 7, \mu_2 = 18, \mu_3 = 26$$

This is different from previous result.
Hence proved



s) a) 0

b) 0

c) 0

d) 0.5

e) 1

f) 1

g) 0.5

h) 0.5

i) 0.8

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